Aircraft External Cleaning – General Data

1. External Cleaning

A. Washing Aircraft Exterior

Equipment and Materials:

- Cleaner MIL–C–25769
- Solvent MIL–C–25179
- Solvent P–D–680 (AMS 3160)
- Isopropanol
- Aliphatic Naphtha TT–N–95, Type II
- Brush, Mop or Sponge
- Cheesecloth

1) General

a) Frequent washing of the aircraft should be carried out to maintain its appearance and to minimize corrosion. The painted area of the aircraft should be polished at periodic intervals to remove chalking of paint and to restore its gloss.

b) Flammable cleaners and solvents should not be used in areas where combustion may occur.

c) To prevent corrosion, do not use high pressure spraying of cleaning detergents on units having holes which drain into internal cavities.

d) Lubricate aircraft components, e.g. pivot points and accessible bearings whenever necessary after washing.

2) Preparation for Cleaning

a) Mask all areas where contact with, or penetration of, cleaning concentrate is to be avoided. To include windshields, windows, wheels, pitot heads, static vents, all top surface openings, doors, nacelle and nose assembly fairings.

b) Use an alkaline water base cleaner Specification MIL–C–25769 diluted in the ratio of one part cleaner to ten parts fresh water for lightly soiled conditions. For moderately soiled conditions, dilute one part cleaner to five to eight parts water.

3) Application of Cleaner

a) Cleaner can be applied by spraying, mop, sponge or brush. Allow cleaner to remain on surface for approximately five minutes and agitate with a brush or mop, rinse thoroughly with copious amounts of clean (potable) water.

Note Do not allow cleaner to dry before rinsing, as less effective cleaning will result. Do not apply the cleaner to excessively large areas which cannot be brushed and thoroughly rinsed before the cleaner dries on the surface.

b) Where electrical equipment is present, limit cleaning wherever possible to wiping clean using cloths moistened with solvent. Specification P–D–680 (AMS3160). Wheels or related brake parts should be hand cleaned using alkaline cleaners or solvents as appropriate.

c) When alkaline cleaning concentrate is not available, solvent emulsion compound Specification MIL–C–25179 may be used, in dilution of one part compound to nine parts cleaning solvent Specification P–D–680 (AMS3160). Procedures are the same as with alkaline water based cleaner but extreme care must be exercised to prevent solution coming into contact with rubber components such as hoses, tires, seals, etc.

d) On completion of cleaning operations, ensure that all masking applied in Para 1.A.2).a), is removed.

B. Cleaning Composite Components

1) Doors, Nose Baggage Compartment, Lower Aft Nacelle Fairing and Fwd/Aft Gallery Fairings. Refer to manufacture's instructions for cleaning procedures for the radome.

2) Apply small amount of solvent to a clean wiping cloth.

Note Apply the minimum amount of solvent, Isopropyl Alcohol, IPA or Plastic Prep/Cleaner eg. Plas-Stick 2319s by Dupont when solvent cleaning.

3) Wipe the composite components with the solvent dampened wiping cloth.

4) Wipe dry with another clean wiping cloth **before** the solvent evaporates.

Note It is important to dry the surface immediately after cleaning because if the solvent evaporates before it is wiped up, oil and grease residue will be left on the clean surface.

5) If more cleaning is required, repeat Para 1.B.2) and Para 1.B.4).

C. Cleaning Aircraft Windows and Windshields

CAUTION DO NOT USE ANY ABRASIVE MATERIALS, STRONG ACIDS, METHANOL OR METHYL ETHYL KETONE.

1) Remove excessive amounts of dirt and other substances from the glass surfaces with clean water applied with a sponge or soft cloth.

2) Wash gently with mild soap and water, or a 50/50 solution of isopropanol and water or aliphatic naphtha TT–N–95, Type II.

3) Rinse thoroughly using clean water and dry using clean, soft cloth or sponge; use a straight rubbing motion.

D. Cleaning Aircraft Propellers

CAUTION DO NOT ALLOW SOLVENT TO CONTACT RUBBER, PLASTIC OR FIBERGLASS PARTS.

1) Clean external metallic surfaces with solvent P–D–680 (AMS3160D) or equivalent. Apply solvent sparingly with a soft cloth or soft brush.

2) Remove solvent by thoroughly rinsing with clean water and wipe dry with a soft cloth.

3) Clean rubber, plastic and fiberglass surfaces with mild soap and water applied with a soft cloth.

4) Rinse with clean water and dry with soft cloth.